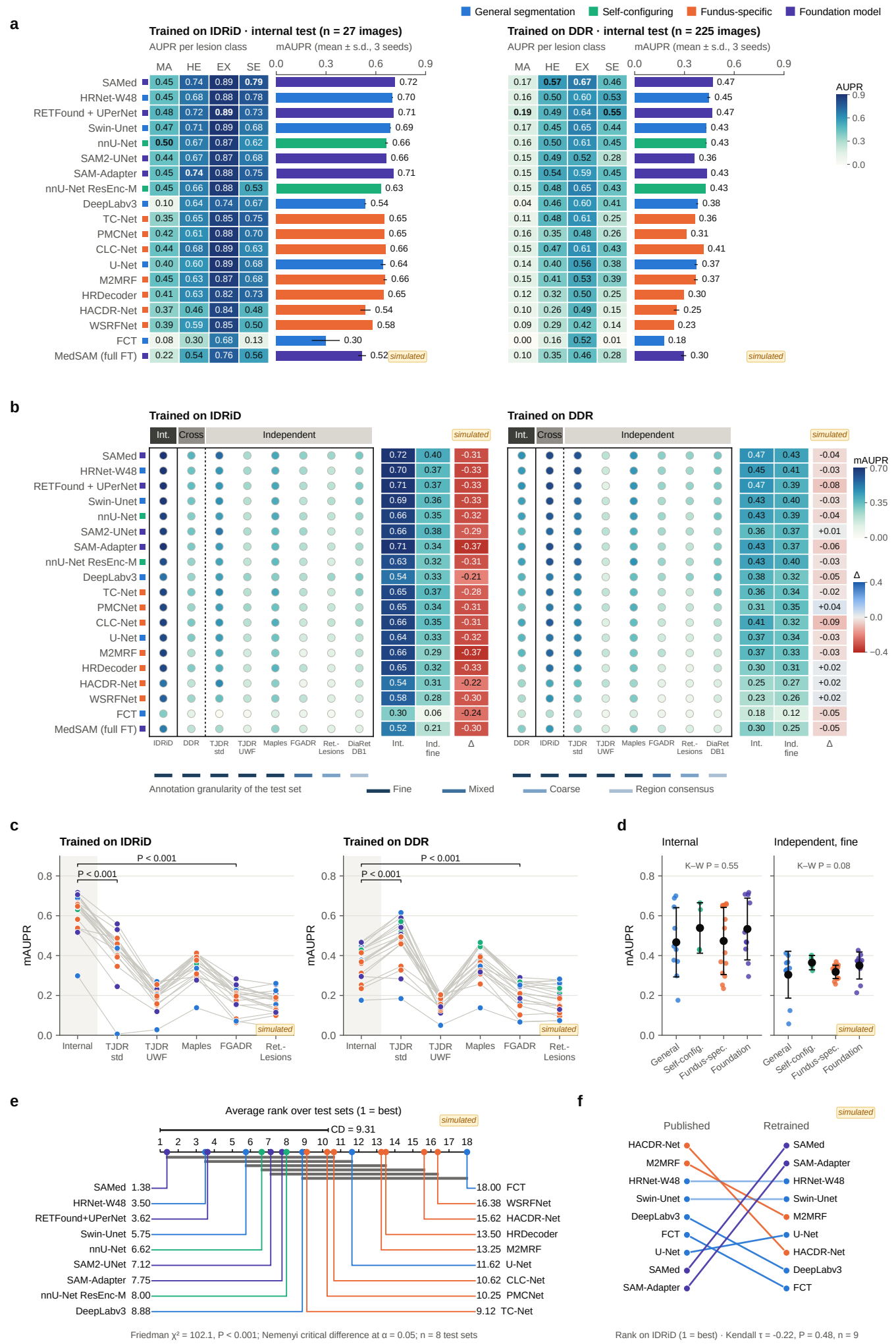

General segmentation
Self-configuring
Fundus-specific
Foundation model
a
Trained on IDRiD · internal test (n = 27 images)
AUPR per lesion class
mAUPR (mean ± s.d., 3 seeds)
MA HE EX SE
SAMed 0.45 0.74 0.89 0.79 0.72
HRNet-W48 0.45 0.68 0.88 0.78 0.70
RETFound + UPerNet 0.48 0.72 0.89 0.73 0.71
Swin-Unet 0.47 0.71 0.89 0.68 0.69
nnU-Net 0.50 0.67 0.87 0.62 0.66
SAM2-UNet 0.44 0.67 0.87 0.68 0.66
SAM-Adapter 0.45 0.74 0.88 0.75 0.71
nnU-Net ResEnc-M 0.45 0.66 0.88 0.53 0.63
DeepLabv3 0.10 0.64 0.74 0.67 0.54
TC-Net 0.35 0.65 0.85 0.75 0.65
PMCNet 0.42 0.61 0.88 0.70 0.65
CLC-Net 0.44 0.68 0.89 0.63 0.66
U-Net 0.40 0.60 0.89 0.68 0.64
M2MRF 0.45 0.63 0.87 0.68 0.66
HRDecoder 0.41 0.63 0.82 0.73 0.65
HACDR-Net 0.37 0.46 0.84 0.48 0.54
WSRFNet 0.39 0.59 0.85 0.50 0.58
FCT 0.08 0.30 0.68 0.13 0.30
MedSAM (full FT) 0.22 0.54 0.76 0.56 0.52
simulated
Trained on DDR · internal test (n = 225 images)
AUPR per lesion class
mAUPR (mean ± s.d., 3 seeds)
MA HE EX SE
0.17 0.57 0.67 0.46 0.47
0.16 0.50 0.60 0.53 0.45
0.19 0.49 0.64 0.55 0.47
0.17 0.45 0.65 0.44 0.43
0.16 0.50 0.61 0.45 0.43
0.15 0.49 0.52 0.28 0.36
0.15 0.54 0.59 0.45 0.43
0.15 0.48 0.65 0.43 0.43
0.04 0.46 0.60 0.41 0.38
0.11 0.48 0.61 0.25 0.36
0.16 0.35 0.48 0.26 0.31
0.15 0.47 0.61 0.43 0.41
0.14 0.40 0.56 0.38 0.37
0.15 0.41 0.53 0.39 0.37
0.12 0.32 0.50 0.25 0.30
0.10 0.26 0.49 0.15 0.25
0.09 0.29 0.42 0.14 0.23
0.00 0.16 0.52 0.01 0.18
0.10 0.35 0.46 0.28 0.30
simulated
AUPR
0.9
0.6
0.3
0.0
b
Trained on IDRiD
Int. Cross Independent
IDRiD DDR TJDR std TJDR UWF Maples FGADR Ret.-Lesions DiaRet DB1 Int. Ind. fine Δ
SAMed 0.72 0.40 -0.31
HRNet-W48 0.70 0.37 -0.33
RETFound + UPerNet 0.71 0.37 -0.33
Swin-Unet 0.69 0.36 -0.33
nnU-Net 0.66 0.35 -0.32
SAM2-UNet 0.66 0.38 -0.29
SAM-Adapter 0.71 0.34 -0.37
nnU-Net ResEnc-M 0.63 0.32 -0.31
DeepLabv3 0.54 0.33 -0.21
TC-Net 0.65 0.37 -0.28
PMCNet 0.65 0.34 -0.31
CLC-Net 0.66 0.35 -0.31
U-Net 0.64 0.33 -0.32
M2MRF 0.66 0.29 -0.37
HRDecoder 0.65 0.32 -0.33
HACDR-Net 0.54 0.31 -0.22
WSRFNet 0.58 0.28 -0.30
FCT 0.30 0.06 -0.24
MedSAM (full FT) 0.52 0.21 -0.30
simulated
Trained on DDR
Int. Cross Independent
DDR IDRiD TJDR std TJDR UWF Maples FGADR Ret.-Lesions DiaRet DB1 Int. Ind. fine Δ
0.47 0.43 -0.04
0.45 0.41 -0.03
0.47 0.39 -0.08
0.43 0.40 -0.03
0.43 0.39 -0.04
0.36 0.37 +0.01
0.43 0.37 -0.06
0.43 0.40 -0.03
0.38 0.32 -0.05
0.36 0.34 -0.02
0.31 0.35 +0.04
0.41 0.32 -0.09
0.37 0.34 -0.03
0.37 0.33 -0.03
0.30 0.31 +0.02
0.25 0.27 +0.02
0.23 0.26 +0.02
0.18 0.12 -0.05
0.30 0.25 -0.05
simulated
mAUPR
0.70
0.35
0.00
Δ
0.4
0.0
−0.4
Annotation granularity of the test set
Fine
Mixed
Coarse
Region consensus
c
Trained on IDRiD
Trained on DDR
P < 0.001
P < 0.001
mAUPR
0.8
0.6
0.4
0.2
0.0
Internal
TJDR std
TJDR UWF
Maples
FGADR
Ret.-Lesions
simulated
d
Internal
Independent, fine
K–W P = 0.55
K–W P = 0.08
mAUPR
General
Self-config.
Fundus-spec.
Foundation
simulated
e
Average rank over test sets (1 = best)
CD = 9.31
simulated
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
SAMed 1.38
HRNet-W48 3.50
RETFound+UPerNet 3.62
Swin-Unet 5.75
nnU-Net 6.62
SAM2-UNet 7.12
SAM-Adapter 7.75
nnU-Net ResEnc-M 8.00
DeepLabv3 8.88
18.00 FCT
16.38 WSRFNet
15.62 HACDR-Net
13.50 HRDecoder
13.25 M2MRF
11.62 U-Net
10.62 CLC-Net
10.25 PMCNet
9.12 TC-Net
Friedman χ² = 102.1, P < 0.001; Nemenyi critical difference at α = 0.05; n = 8 test sets
f
Published
Retrained
simulated
HACDR-Net
M2MRF
HRNet-W48
Swin-Unet
DeepLabv3
FCT
U-Net
SAMed
SAM-Adapter
SAMed
SAM-Adapter
HRNet-W48
Swin-Unet
M2MRF
U-Net
HACDR-Net
DeepLabv3
FCT
Rank on IDRiD (1 = best) · Kendall τ = -0.22, P = 0.48, n = 9